Power Markets Casebook

v0.36.3

> Applied decision cases for turning market, grid, contract, and portfolio concepts into professional screens and memos.

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v0.36.3

The Casebook gives bounded commercial exercises. Each case asks for a concrete artifact: a quote stack, duration screen, site screen, partner scorecard, product memo, or operating-model choice.

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Methodology

These notes were built with a learning method designed for entering technical domains quickly: gather strong primary sources, identify the core guiding questions, work back to first principles, and build intuition through orders of magnitude, scaling relationships, and worked decision frames.

The workflow was accelerated by a custom AI research and knowledge-management system I built for semantic retrieval, source organization, drafting, and the editorial loops.

These notes are evolving. As I learn more and receive feedback, I keep improving the material.

Feedback is very welcome. When reading a case, select a relevant passage and click 'Send feedback' to send me your input on the spot.

How To Use The Casebook

Pick a case when the corresponding lecture idea feels familiar enough to use. If the answer feels hand-wavy, return to the linked lectures and rebuild the artifact.

- Read the decision question before the scenario data.
- Build the requested artifact before reading the worked solution.
- Compare your recommendation to the assumption flip.

Casebook Structure

The cases move from commercial screens to operating-model choices. Each case is standalone, and together they test whether the concepts can produce usable work.

Casebook

Applied Decision Practice

The cases give bounded practice in screens, memos, scorecards, product tests, and operating-model choices.

- Use the given facts before adding judgment.
- Find the binding constraint and the first assumption that changes the answer.
- Produce a reusable artifact that records the decision logic.

Start Case 1 **Q&A Overview PDF**

Glossary

An alphabetical reference for the terms, acronyms, market roles, and technical objects used throughout the notes.

- Use it for quick recall.
- Return to the relevant lecture when the term needs mechanism, economics, or context.

Open Glossary

References

References

Books, reports, official pages, and selected practitioner analyses used across the notes.

- Market figures and rule-specific details include dates where timing matters.

Open References

Applied Cases

Case 1. Shaped Solar-Storage Power Purchase Agreement Quote

References: [1] [2] [3] [4] [5] [6]

A buyer asking for “solar plus storage” is usually asking for several things at once: price certainty, evidence of clean supply, a volume shape, and someone to carry the mismatch between production and consumption. The seller can accept that mismatch, sell a simpler product, or price the shape explicitly. The work product is a quote stack and a decision memo that make the residual obligation visible before the term sheet turns it into risk.

Decision Question

Should a renewable independent power producer (IPP) offer a shaped solar-plus-battery energy storage system (BESS) power purchase agreement (PPA), a simpler as-produced PPA, or no-bid?

Situation

A corporate buyer wants a 10-year clean-power product for a stable industrial load. The buyer prefers a flat hourly shape because its procurement team wants budget certainty and less exposure to evening prices.

The seller owns a solar project and can add a co-located battery. The commercial team asks for a quote that is attractive to the buyer, acceptable to investment committee, and explicit about who owns residual risk.

Scenario Data

These are scenario numbers for the case, not market quotes.

Item	Value
Buyer annual demand	100 GWh/year
Product requested	100 gigawatt-hours (GWh)/year flat profile, about 11.4 megawatts (MW) average
Solar project output	90 GWh/year

Item	Value
Co-located BESS	25 MW / 50 megawatt-hours (MWh)
Battery round-trip efficiency	88%
As-produced solar offer floor	45 EUR/MWh
Buyer maximum acceptable shaped price	68 EUR/MWh
Stress firming premium in hard hours	120 EUR/MWh
Residual hard-hour volume after solar+ BESS	20 MWh/h
Number of hard hours in stress case	250 h/year

For the base quote, use these illustrative quote-stack inputs. They are scenario assumptions for the exercise, not market benchmarks.

Quote component	EUR/MWh
Raw as-produced clean energy	45.0
Shape and firming cost	9.0
Expected imbalance and forecast error	2.0
Grid, tariff, and settlement administration	1.5
Certificates and evidence handling	1.0
Credit and tenor allowance	1.5
Battery opportunity cost	3.0
Risk margin	2.0

Product Boundary To Price

Before quoting a number, the seller needs to define the product boundary. A shaped PPA bundles clean MWh with settlement, evidence, and fallback obligations.

Term	Case Boundary	Why It Matters
Quantity	about 11.4 MW flat hourly profile, 100 GWh/year commercial target	fixes the shape being sold
Physical source	seller solar output plus co-located BESS where available	separates asset-backed delivery from market firming
Settlement	fixed shaped price against the agreed hourly profile	determines who pays when production and load differ
Residual firming	seller covers expected residuals; hard-hour tail must be capped, indexed, or passed through	prevents an uncapped short-volatility position
Imbalance	balance responsible party (BRP)-managed, with agreed data and performance reporting	connects contract economics to operational evidence
Certificates and matching evidence	defined certificate vintage, geography, delivery period, and data package	avoids selling a clean-energy claim without proof rules
Dispatch rights	seller/optimizer retains rights needed to deliver the shape and manage the BESS	stops the same flexibility being promised elsewhere
Data and audit rights	hourly meter data, certificate evidence, residual procurement records,	makes the product testable after delivery

Term	Case Boundary	Why It Matters
	dispatch logs, and settlement files are deliverable	
Credit and collateral	buyer credit support and seller exposure limit stated in the term sheet	prevents tenor and collateral from being buried in the energy price

Tasks

1. Build a quote stack for the shaped product.
2. Identify which risks remain with the seller.
3. Decide whether to offer the shaped product, offer only as-produced energy, or no-bid.
4. State the first assumption that would change the decision.

Worked Solution

The quote stack must start with the energy the asset can naturally provide, then add the cost of reshaping that energy into the buyer's requested profile. In this scenario, the buyer asks for `100 GWh/year` flat while the solar project produces `90 GWh/year`; the annual clean-energy shortfall and any certificate-backed replacement supply have to be priced separately from hard-hour firming. The quote is usable only if the term sheet assigns the residual mismatch to a party, a price, a cap, or a reopener.

Component	EUR/MWh	Risk Owner If Included In Fixed Price
Raw as-produced clean energy	45.0	Seller
Shape and firming cost	9.0	Seller
Expected imbalance and forecast error	2.0	Seller / BRP contract
Grid, tariff, and settlement administration	1.5	Seller
Certificates and evidence handling	1.0	Seller
Credit and tenor allowance	1.5	Seller

Component	EUR/MWh	Risk Owner If Included In Fixed Price
Battery opportunity cost	3.0	Seller
Risk margin	2.0	Seller
Indicative shaped quote	**65.0**	

At 65 EUR/MWh, the product sits below the buyer’s stated 68 EUR/MWh ceiling. The tail risk still needs an explicit boundary.

Numerical Intuition

The stress firming cost is:

```
20 MWh/h x 250 h/year x 120 EUR/MWh = 600,000 EUR/year
```

The fixed risk margin in the quote is:

```
100,000 MWh/year x 2 EUR/MWh = 200,000 EUR/year
```

The quoted margin falls short of the illustrated stress case. The product can still be commercial if the contract caps the seller’s hard-hour procurement risk at a fixed price.

Decision Memo

Recommendation: offer the shaped product at 65 EUR/MWh only with a residual-firming collar.

The seller should own expected shaping and imbalance risk inside the quoted stack. The buyer should accept either a pass-through, cap, index, or renegotiation trigger for residual hard-hour firming beyond the agreed collar. Without that boundary, the product embeds a short volatility position inside the PPA.

The investment-committee test is whether the price, cap, credit terms, evidence package, data rights, and dispatch rights describe the same product. If the buyer wants the seller to carry unlimited residual firming, the quote must move up or the product must narrow.

The as-produced fallback should remain on the table at 45 EUR/MWh. It is less valuable to the buyer, but it avoids making a fixed-price promise that the asset cannot physically support under stress.

Assumption Flip

The recommendation changes if the residual hard-hour volume rises from 20 MWh/h to 35 MWh/h.

```
35 MWh/h x 250 h/year x 120 EUR/MWh = 1,050,000 EUR/year
```

At that point the seller either needs a much higher fixed price, a portfolio hedge, a third-party firming partner, a softer shape, or a no-bid decision.

Failure Mode To Avoid

Approve the fixed shaped quote because the expected quote stack is below the buyer's ceiling, then leave hard-hour residuals to be absorbed by the portfolio without a cap, index, or reopener.

That decision leaves the portfolio carrying an unbounded residual exposure. The product promises shape, timing, settlement, and fallback procurement alongside annual clean energy.

Links Back To The Lectures

- Load Is Not Energy
- Solar Production
- PPA Risk Allocation
- PPA Structure Catalog
- PPA Pricing
- BESS Revenue Stacks
- Dispatch Optimization

Case 2. The 1h, 2h, Or 4h Battery Decision

References: [7] [8] [9] [10] [11] [12] [5]

Battery duration is an investment claim: the next MWh of energy capacity must earn more than it costs after efficiency, degradation, tariffs, grid rights, and contract obligations. This case asks the duration question as an investment screen rather than as a technology preference.

Decision Question

For a constrained solar site, should the project build a 1-hour, 2-hour, or 4-hour battery?

Situation

A solar project has secured land and a grid position. The grid connection can export the solar plant, but adding storage requires choosing a battery duration before the PPA and optimizer contracts are finalized.

The commercial team wants a simple answer. The answer depends on how scarcity, grid access, contract value, warranty limits, and insurance limits change from the first hour to later hours.

Scenario Data

These are scenario assumptions for the case.

Item	Value
Solar project size	80 MWp
Grid export limit	60 MW
Battery power rating	30 MW
Candidate durations	1h, 2h, 4h
Carrying charge used for screen	10% of incremental capex/year

Item	Value
Round-trip efficiency	88%
Degradation allowance	included in net margin
Insurance policy	scenario assumes the modeled operating plan is insurable; if a policy caps cycles or duty cycle below the modeled plan, rerun the screen under that cap

Incremental Block	Incremental Energy	Incremental Capex	Annual Hurdle
First hour	30 MWh	5.0 million EUR	0.50 million EUR/year
Second hour	30 MWh	3.5 million EUR	0.35 million EUR/year
Third and fourth hours	60 MWh	7.0 million EUR	0.70 million EUR/year

Incremental Block	Usable Cycles/Year	Net Value Per MWh Moved	Extra Curtailment /Contract Value
First hour	310	55 EUR/MWh	0.05 million EUR/year
Second hour	240	45 EUR/MWh	0.15 million EUR/year
Third and fourth hours	120	25 EUR/MWh	0.07 million EUR/year

Duration Gates

The duration decision needs more than an arbitrage calculation. Each incremental hour must pass the commercial, technical, and contractual gates that make the model value collectible.

Gate	What To Check	Failure Signal
Marginal value	the next MWh of energy capacity clears its own hurdle	average spread from the first hours is applied to all hours
Cycle, warranty, and insurance budget	modeled throughput, average state of charge, C-rate or charge/discharge rate, temperature envelope, data-retention obligation, degradation allowance, warranty terms, and insurance policy are mutually compatible	optimizer earns value by consuming life, evidence, or coverage headroom not priced in the case
Grid import and export rights	connection agreement permits the charging and discharging pattern	battery cannot access the hours that justify the duration
Product eligibility	extra duration qualifies for the intended reserve, firming, or PPA product	extra MWh exists physically but earns no contracted premium
Control rights	dispatch rights remain with the party whose economics justify the investment	PPA, optimizer, and reserve obligations all assume priority
Augmentation option	site layout, transformers, inverters, and permits leave room for later expansion	management must choose all duration now under weak evidence

DNV’s storage-warranty review makes the second gate concrete. A capacity warranty may care about usage/cycling degradation, calendar degradation, temperature, average state of charge, C-rate, data storage, and repair response. The case therefore treats “cycles/year” as a commercial shorthand. Before live use, replace it with the warranty-compatible operating envelope and the actual insurance-policy limits.

Tasks

1. Calculate the annual value of each duration block.
2. Compare each block with its annual hurdle.
3. Recommend 1h, 2h, or 4h.
4. State what evidence would change the decision.

Worked Solution

The first hour value:

```
30 MWh x 310 cycles/year x 55 EUR/MWh + 50,000 EUR/year
= 561,500 EUR/year
```

The first hour clears its 500,000 EUR/year hurdle by about 61,500 EUR/year.

The second hour value:

```
30 MWh x 240 cycles/year x 45 EUR/MWh + 150,000 EUR/year
= 474,000 EUR/year
```

The second hour clears its 350,000 EUR/year hurdle by about 124,000 EUR/year.

The third and fourth hour value:

```
60 MWh x 120 cycles/year x 25 EUR/MWh + 70,000 EUR/year
= 250,000 EUR/year
```

The third and fourth hours miss their 700,000 EUR/year hurdle by about 450,000 EUR/year.

Numerical Intuition

The best duration is the last block whose marginal value exceeds its marginal cost.

Option	Annual Value	Annual Hurdle	Margin To Hurdle
1h	0.56 million EUR/year	0.50 million EUR/year	+0.06 million EUR/year
2h	1.04 million EUR/year	0.85 million EUR/year	+0.19 million EUR/year
4h	1.29 million EUR/year	1.55 million EUR/year	-0.26 million EUR/year

The 4-hour option creates more gross value than the 2-hour option, but it destroys value at the margin in this scenario.

Decision Memo

Recommendation: build the 2-hour battery, preserve design optionality for later augmentation, and do not commit to 4 hours until the site has stronger evidence of longer-duration scarcity or contracted value.

The binding constraint is the weak marginal value of hours three and four. The 2-hour asset captures the strongest short-duration price and curtailment value while avoiding a large capital block with too few high-margin cycles.

The investment decision should approve the 2-hour build and require the design team to preserve practical augmentation options where the cost is modest: space, cable routes, transformer headroom, inverter/control compatibility, and permitting language. Preserving those options keeps the project from paying today for duration that the evidence does not yet support.

The commercial team should ask for six pieces of evidence before revisiting 4 hours:

- a contract that pays for longer sustained delivery
- tariff or grid rules that reward longer import/export management
- historical or forecast price data showing enough multi-hour scarcity events after degradation and fees
- warranty, insurance, and augmentation terms showing that the value can be captured without consuming battery life or coverage headroom faster than priced
- battery management system data-access and storage arrangements strong enough to support warranty, availability, and safety disputes
- dispatch-control rights showing how PPA, reserve, merchant, and emergency uses are prioritized against the same state of charge

Assumption Flip

The 4-hour decision changes if the third and fourth hours can earn 70 EUR/MWh net value for 160 cycles/year.

```
60 MWh x 160 cycles/year x 70 EUR/MWh + 70,000 EUR/year
= 742,000 EUR/year
```

It can also change in the other direction if an insurance policy caps eligible operation below the modeled cycle need. In that case, replace the modeled useful cycles with the policy-compatible cycle count before comparing duration blocks. The project may still proceed, but only after the owner obtains different coverage, changes the operating plan, or prices the lost optionality.

That clears the 700,000 EUR/year hurdle. The case for 4 hours depends on the frequency and value of longer scarcity windows, and on whether the operating plan remains compatible with warranty, insurance, tariff, and control-right constraints.

Failure Mode To Avoid

Choose four hours because the incremental capex per MWh is lower than the first two hours, then apply the average short-duration spread to all four hours.

That decision uses the economics of the valuable first hours to justify weaker later hours. The third and fourth hours need their own scarcity evidence, contract value, tariff benefit, or portfolio hedge value.

Links Back To The Lectures

- Frequency And Reserves
- Battery Storage And Time Shifting
- BESS Revenue Stacks
- Battery Duration Selection
- Battery Degradation, Warranties, And Operating Constraints
- Storage Duration And Long-Duration Flexibility

Case 3. Grid-Constrained Site Screen

References: [11] [13] [14] [15] [16] [17] [18]

Solar development often begins with land and irradiation. Project value often ends with grid access, tariffs, congestion, curtailment, and what the connection agreement actually permits. The work product is a grid-aware investment screen for three attractive sites.

Decision Question

Which site should the IPP prioritize when irradiation, grid rights, tariffs, and portfolio fit point in different directions?

Situation

The development team has three candidate sites for a new solar-plus-storage project. All three can be developed, but only one can receive near-term investment resources.

The highest-yield site must still be tested against grid economics and commercial optionality.

Scenario Data

These are scenario assumptions.

Metric	Site A	Site B	Site C
Solar size	100 MWp	90 MWp	110 MWp
Expected PV yield	1,060 kWh/kWp/year	1,000 kWh/kWp/year	1,030 kWh/kWp/year
Earliest firm connection	2030	2028	2028
Export limit	65 MW	80 MW	55 MW
Expected curtailment	5%	1%	8%

Metric	Site A	Site B	Site C
Grid/tariff/loss cost	8 EUR/MWh	4 EUR/MWh	11 EUR/MWh
BESS import allowed	conditional	yes	no
Portfolio zone fit	weak	strong	medium
Permitting complexity	medium	medium	low

Assume the merchant value before grid costs is 48 EUR/MWh for all sites. Ignore capex differences for the first screen.

Tasks

1. Calculate net annual MWh after curtailment.
2. Calculate net value per MWh after grid/tariff/loss cost.
3. Estimate annual gross margin before capex and opex.
4. Add a grid-rights and timing screen.
5. Recommend the priority site.

Worked Solution

Annual MWh before curtailment:

Site	Calculation	Gross Output
A	100,000 kWp x 1,060 kWh/kWp	106,000 MWh/year
B	90,000 kWp x 1,000 kWh/kWp	90,000 MWh/year
C	110,000 kWp x 1,030 kWh/kWp	113,300 MWh/year

After curtailment:

Site	Net MWh/year
A	100,700
B	89,100
C	104,236

Net value per MWh:

Site	Value Before Grid Cost	Grid/Tariff/Loss Cost	Net Value
A	48 EUR/MWh	8 EUR/MWh	40 EUR/MWh
B	48 EUR/MWh	4 EUR/MWh	44 EUR/MWh
C	48 EUR/MWh	11 EUR/MWh	37 EUR/MWh

Annual gross margin before capex and opex:

Site	Net MWh/year	Net Value	Annual Gross Margin
A	100,700	40 EUR/MWh	4.03 million EUR/year
B	89,100	44 EUR/MWh	3.92 million EUR/year
C	104,236	37 EUR/MWh	3.86 million EUR/year

Grid Rights Screen

Site	Grid And Commercial Signal	Investment Meaning
A	highest base margin, but 2030 firm connection, conditional BESS import, weak zone fit	value is delayed and the storage strategy is not yet firm

Site	Grid And Commercial Signal	Investment Meaning
B	lower output, but 2028 connection, low curtailment, full BESS import, strong zone fit	lower headline MWh but better executable optionality
C	high output, no BESS import, high curtailment and grid cost	looks large on MWp but weak for shaped products and flexibility

The tariff and connection question includes EUR/MWh, reserved capacity, geography, limited access, and producer payments. Nordic tariff work increasingly separates these items. A site screen should therefore ask what capacity is being reserved, when it is firm, whether limited or interruptible access is acceptable, and whether import rights let the battery charge when the commercial case requires it.

Numerical Intuition

The top-line spread between sites is small after grid effects.

```
Site A gross margin - Site B gross margin = about 0.11 million EUR/year
```

That uplift is too small to dominate a two-year connection delay, weaker portfolio fit, conditional battery import rights, and higher grid uncertainty.

The undiscounted value delayed by choosing Site A over a near-term 2028 site is roughly:

```
2 years x 4.03 million EUR/year = 8.06 million EUR of gross margin timi
```

Use this as a screen, rather than a project net present value (NPV), to prevent a small annual yield advantage from hiding a large timing and optionality penalty.

Decision Memo

Recommendation: prioritize Site B.

Site A has the highest simple annual gross margin, but Site B has the best decision quality. It connects earlier, has lower curtailment, lower grid cost, stronger portfolio fit, and full BESS import rights. Those features make it more useful for shaped PPAs, reserve participation, and portfolio balancing.

The first diligence request should be the grid-rights evidence pack: firm connection date, export capacity, import capacity, connection contribution, reinforcement cost, securities or deposits, queue milestones, curtailment rules, capacity-tariff exposure, producer-payment geography, limited-access terms, metering boundary, and whether BESS operation is allowed under the same connection agreement or needs a separate approval.

Site A should remain in development if the connection timing improves or if a buyer values the location enough to pay for the grid uncertainty. Site C is not a priority unless its low permitting complexity creates a materially faster path than assumed.

Assumption Flip

The recommendation changes if Site A obtains firm 2028 connection and full BESS import rights.

Then the main penalty is grid cost rather than timing and optionality. Site A's higher output may justify prioritization, especially if the portfolio already has enough exposure in Site B's zone.

Failure Mode To Avoid

Rank the sites by base-case P50 generation value, then treat connection timing, import rights, curtailment, metering boundary, and tariff exposure as diligence items.

That decision puts the grid outside the economics even though the grid determines when the MWh is deliverable, what the battery can do, and which commercial products remain feasible.

Links Back To The Lectures

Case 4. BESS Revenue Stacks Under Saturation

References: [19] [20] [21] [22] [7] [9] [12] [23] [5] [24]

A battery revenue stack is a set of coupled obligations on the same MW, MWh, state of charge, grid rights, cycle budget, insurance policy, and control system. When a service saturates, test which obligations remain valuable after opportunity cost and downside are counted.

Decision Question

Should the asset owner contract part of the battery's flexibility, keep it merchant, or reserve optionality while the revenue stack evolves?

Situation

A 50 MW / 100 MWh battery can participate in energy arbitrage and reserve products. A utility offers a five-year floor contract for part of the asset. An optimizer offers full merchant dispatch with no guaranteed payment. A customer team asks whether part of the battery should be reserved for shaped PPA delivery.

The project cannot sell the same flexibility three times.

An optimizer can schedule within the conflict; it cannot remove the conflict. DNV's hybrid-storage optimizer material is useful here because it frames dispatch as a constrained problem over market products, loss trees, interconnection limits, cycling and throughput requirements, warranty strategies, and capacity maintenance. A revenue stack is credible only if the same constraint set governs every product in the stack.

Scenario Data

These are scenario assumptions.

Item	Value
Battery	50 MW / 100 MWh
Round-trip efficiency	88%

Item	Value
Expected merchant stack value	1.70 million EUR/year
Downside merchant stack value	0.70 million EUR/year
Floor contract fixed payment	0.90 million EUR/year
Expected upside retained under floor	0.30 million EUR/year
Shaped PPA flexibility premium	0.75 million EUR/year
Merchant value left after reserving PPA capacity	0.80 million EUR/year
Lender-counted share of pure merchant value	20% scenario assumption
Lender-counted share of fixed floor	100% scenario assumption
Insurance cycle limit	scenario assumes no binding limit beyond the modeled cycle budget; verify policy terms before using the stack

Tasks

1. Compare expected value and financeable value.
2. Identify which option has the strongest downside protection.
3. State the opportunity cost of reserving capacity for a shaped PPA.
4. Recommend an allocation.

Contracting Routes

Recent BESS offtake discussions use several structures rather than one generic “battery revenue” line. The case reduces them to four routes the investment committee can compare.

Route	What Is Sold	Control-Rights Consequence	When It Fits
Full merchant optimization	optimizer pursues energy, reserve, and imbalance opportunities	asset owner keeps upside and downside; lender credit is limited	owner accepts volatility and has strong controls
Floor or revenue-share contract	floor, toll, revenue swap, or minimum guaranteed payment	some upside and dispatch discretion are transferred; availability, reporting, termination, and exclusivity terms matter	project needs financeability or downside protection
Partial hybrid PPA	defined battery flexibility supports a customer profile, residual remains merchant or contracted separately	part of the battery is reserved for the PPA product	customer premium exceeds opportunity cost and risk boundary is clear
Combined hybrid PPA	renewable energy and BESS are packaged into one customer product	larger share of flexibility is locked to the customer	buyer values firmness enough to pay for it and evidence/control rights are explicit

Worked Solution

Route	Expected Value	Downside Value	Financeable Value In This Scenario	Main Constraint
Full merchant	1.70 million EUR/year	0.70 million EUR/year	0.34 million EUR/year	Volatile and weakly financeable

Route	Expected Value	Downside Value	Financeable Value In This Scenario	Main
Floor contract	1.20 million EUR/year	0.90 million EUR/year	0.90 million EUR/year	Gives up some upside
Shaped PPA plus residual merchant	1.55 million EUR/year	depends on residual firming	0.75 million EUR/year plus residual credit	Ties up control rights

The full merchant route has the highest expected value, but the lowest financeable value. The floor contract has lower expected value but stronger downside and lender readability. The shaped PPA route is attractive only if the PPA premium is high enough to compensate for lost merchant optionality.

Numerical Intuition

The opportunity cost of using battery capacity for the shaped PPA is:

```
full merchant expected value − residual merchant value
= 1.70 million − 0.80 million
= 0.90 million EUR/year
```

The shaped PPA pays 0.75 million EUR/year for that flexibility. On expected value alone, it is short by 0.15 million EUR/year before considering customer value, contract tenor, and financeability.

Decision Memo

Recommendation: contract a defined tranche of the asset under a floor-like structure and keep a meaningful residual slice merchant. The tranche must state MW, MWh, state-of-charge band, cycle budget, priority rights, and reporting obligations. Allocate battery capacity to the shaped PPA only if the PPA premium rises or the PPA materially improves project financeability.

The binding constraint is control-right allocation. Every MW reserved for a customer profile is a MW unavailable for reserve capacity, arbitrage, or response to a new market product. The term sheet should therefore state the priority order for reserve obligations, customer shape delivery, merchant arbitrage, degradation limits, project-policy limits, minimum state of charge, availability reporting, termination triggers, and emergency override.

It should also state the evidence layer: battery management system data access, data retention, performance-test method, availability calculation, response/repair clock, and who may override dispatch during safety, grid, or warranty events. Without that evidence layer, the floor provider, optimizer, customer, insurer, and owner can each believe they own the same operating headroom.

The commercial team should price shaped PPA flexibility at its opportunity cost. If the customer cannot pay the opportunity cost, use a narrower shaped PPA plus a separate floor or optimizer agreement for the battery.

The decision pack should show two values side by side: expected value and financeable downside value. A route that maximizes expected merchant upside may still be the wrong first financing structure if it leaves the project with weak debt sizing, high collateral exposure, or no lender-readable revenue floor.

Assumption Flip

The recommendation changes if merchant revenues saturate and the expected merchant stack falls by 40%.

```
1.70 million EUR/year x 60% = 1.02 million EUR/year
```

In that world, recompute the shaped-PPA opportunity cost using the reduced merchant and residual values. The floor becomes more attractive, and the shaped-PPA premium may clear only if the repriced opportunity cost falls below the premium.

Failure Mode To Avoid

Use last year’s reserve revenue, an arbitrage forecast, and a shaped-PPA premium as three additive upside lines, then haircut the total for prudence.

The same failure appears when the stack ignores project-specific policy constraints. If the actual insurance, warranty, lender, or operating documents allow fewer cycles, different duty cycles, or fewer operating modes than the optimizer assumes, part of the stack is unavailable unless the owner obtains approval, changes coverage, or changes operation.

Both versions double-count flexibility. The same MW, MWh, state of charge, control rights, cycle budget, warranty envelope, data evidence, and grid connection must serve every product. Revenue stacking is constrained co-optimization; simple spreadsheet addition double-counts the asset.

Links Back To The Lectures

- Frequency And Reserves
- Battery Storage And Time Shifting
- BESS Revenue Stacks
- Dispatch Optimization
- Valuation Under Uncertainty
- BESS Revenue Saturation

Case 5. Balance-Responsibility And Balancing-Service Performance Audit

References: [25] [19] [26] [27] [28] [29]

A trading or optimization partner can execute the market interface, but the asset owner still needs evidence before it changes fees, mandates, incentives, or partners. The main work is attribution. A monthly revenue gap may come from market conditions, missing data, asset constraints, forecast error, risk limits, credit limits, settlement corrections, or dispatch choices. Partner action is justified only after the evidence separates those causes.

Decision Question

Should the asset owner change partner terms now, require evidence remediation first, or treat the gap as market and constraint noise?

Situation

A renewable IPP outsources balance responsible party (BRP) and balancing service provider (BSP) execution work. The partner sends monthly revenue reports. The asset owner wants to know whether the partner created value, leaked value, or operated within realistic constraints before opening a commercial renegotiation.

The partner contract allows performance review, but the current data annex is weak. Forecast files, bid files, schedule revisions, optimizer recommendations, manual overrides, meter data, activation records, and settlement corrections are delivered inconsistently.

Scenario Data

These are scenario numbers for one month. The revenue row is already net of imbalance, curtailment, and reserve effects. The lower rows are raw evidence inputs for attribution, not extra lines to add or subtract from revenue.

Metric	Actual Partner Result	Feasible Benchmark
Net energy and reserve revenue	5.90m EUR	6.40m EUR
Imbalance cost	−0.42m EUR	−0.31m EUR

Metric	Actual Partner Result	Feasible Benchmark
Curtailed or unavailable energy value	`−0.18m EUR`	`−0.17m EUR`
Data completeness	`92%`	`99% required`
Forecast files delivered on time	`88%`	`98% required`
Bids matching agreed risk limits	`100%`	`100% required`

The feasible benchmark uses only information available before gate closure, actual grid and asset limits, contracted control rights, agreed risk limits, and market products the partner was allowed to access.

Raw attribution pack:

Evidence Item	Observation	Initial EUR Impact
Missing or late data	forecast and bid files missing for several high-spread intervals	`0.12m`
Grid and asset constraints	export limit and outage periods reduced feasible dispatch	`0.08m`
Forecast error inside agreed range	weather-driven production miss within tolerance band	`0.09m`
Risk-limit conservatism	partner stayed inside owner-approved risk limits during volatile hours	`0.10m`
Residual bidding/dispatch gap	unexplained difference after known constraints	`0.11m`

Evidence quality flags:

Required Evidence	Status	Consequence
Forecast versions and timestamps	incomplete	cannot prove what was knowable
Bid submissions and revisions	partial	cannot fully test partner action
Optimizer recommendations	missing for 18% of intervals	cannot separate tool advice from manual override
Asset availability and outages	complete	physical constraints can be attributed
Meter and settlement data	provisional, corrections pending	revenue result may move
Collateral and credit-limit evidence	partial	cannot tell whether otherwise feasible trades were blocked
Contract rights and risk limits	complete	feasible benchmark boundary is known

Tasks

1. Calculate the headline gap.
2. Attribute the gap using the raw evidence.
3. Decide which amount can be used for partner action now.
4. Draft a scorecard and governance recommendation.

Worked Solution

Headline gap:

```
benchmark_revenue - actual_revenue
= 6.40m EUR - 5.90m EUR
= 0.50m EUR
```

Attribution:

Driver	EUR Impact	Governability	Can Support Partner Action Now?
Missing or late data	`0.12m`	shared governance / contract data rights	yes, for data remediation
Grid and asset constraints	`0.08m`	mostly non-controllable	no
Forecast error inside agreed range	`0.09m`	model improvement	no, unless tolerance is renegotiated
Risk-limit conservatism	`0.10m`	owner policy choice	no, unless mandate changes
Residual bidding/dispatch gap	`0.11m`	partner execution candidate	only after evidence file is complete
Total	**`0.50m`**		

The only clean commercial action this month is data remediation. The `0.11m EUR` residual may be partner execution leakage, but the missing forecast, bid, and optimizer files make a fee claim premature.

Numerical Intuition

The headline gap overstates the partner-action amount.

```
residual_bidding_gap_share = 0.11 / 0.50 = 22%
data_and_observability_share = 0.12 / 0.50 = 24%
```

Nearly a quarter of the gap is an evidence problem, and only about a fifth is a candidate execution problem. A strong scorecard prevents the owner from negotiating a `0.50m EUR` accusation when the supported execution claim is closer to `0.11m EUR` and still evidentially weak.

Evidence Required Before Partner Action

The owner should require raw files before using the scorecard for fee clawback, termination, incentive reset, or mandate reduction.

Evidence Layer	Required Files Or Data	What It Separates
Forecast and schedules	forecast versions, schedules, revisions, timestamps, gate-closure state	model error versus late information
Bids and dispatch	submitted bids, accepted bids, activations, optimizer recommendations, manual overrides	partner choice versus market acceptance
Asset and grid constraints	availability, outages, import/export limits, reserve commitments	controllable execution versus physical constraint
Meter and settlement	metered volumes, imbalance settlement, reserve settlement, corrections	economic result versus provisional estimate

Evidence Layer	Required Files Or Data	What It Separates
Contract rights	data rights, control rights, risk limits, permitted products	what the partner was allowed to do
Credit and collateral	collateral calls, credit limits, settlement-bank or counterparty notices	revenue result versus liquidity constraint
Service-level agreement (SLA) and escalation trail	dated exceptions, response times, remediation owners, and repeated defects	one-off error versus governable pattern

This mirrors the logic in Nordic settlement governance: performance and imbalance outcomes depend on reported data, settlement quantities, participant obligations, and corrections. A private partner scorecard should be at least as disciplined about raw evidence.

Scorecard

Use a scorecard that separates result, evidence, and attribution.

Scorecard Area	Month Result	Status	Action
Revenue versus feasible benchmark	`-0.50m EUR`	watch	track with attribution; no fee action on headline gap
Data completeness	`92%` versus `99%` requirement	fail	remediation plan with dated file-delivery obligations
Forecast timeliness	`88%` versus `98%` requirement	fail	daily timestamp feed and exception report
Risk-limit compliance	`100%`	pass	keep current limits until management

Scorecard Area	Month Result	Status	Action
			reviews risk appetite
Constraint attribution	`0.08m EUR`	accepted	exclude from partner execution claim
Residual execution candidate	`0.11m EUR`	unresolved	retest after complete raw files and settlement corrections

The scorecard should be repeated for at least three comparable months before commercial penalties are applied, unless the contract already defines automatic data-delivery breaches.

Decision Memo

Recommendation: do not renegotiate fees or accuse the partner based on the `0.50m EUR` headline gap. Issue a formal data-rights and reporting remediation notice, agree the feasible-benchmark methodology, and put the partner on a three-month evidence watch.

Immediate actions:

- amend or enforce the data annex for forecast, schedule, bid, dispatch, optimizer, meter, activation, and settlement files
- require timestamped delivery and exception reporting
- freeze the feasible-benchmark method before seeing next month's results
- review whether owner-approved risk limits or credit limits are too conservative for volatile periods
- retest residual bidding/dispatch gaps only where evidence is complete

Partner commercial action becomes supportable if, for three comparable months:

- data completeness meets the `99%` requirement
- forecast timeliness meets the `98%` requirement
- settlement corrections are reconciled

- the residual bidding/dispatch gap remains material against an agreed EUR threshold or percentage of monthly benchmark revenue
- the gap occurs in intervals where action was feasible and permitted

Assumption Flip

The recommendation changes if complete raw files show that the partner ignored feasible profitable bids, violated the agreed strategy, or failed to act inside clear market windows while limits and asset constraints allowed action.

In that case, the `0.11m EUR` residual becomes an execution claim rather than an unresolved attribution bucket, and the owner can move to fee adjustment, incentive redesign, tighter mandate, or replacement process.

Failure Mode To Avoid

Use a perfect-hindsight dispatch and attribute the full `0.50m EUR` gap to the partner.

Perfect hindsight can estimate an upper bound, but it cannot govern a BRP/BSP relationship. The governance benchmark must use forecasts, market prices, liquidity, grid limits, asset status, control rights, and risk limits available at the time.

Links Back To The Lectures

- Market Participants And Responsibilities
- Market Clock
- Forecast Error And Imbalance
- Frequency And Reserves
- Forecasting, Error, And Commercial Value
- Trading Partners And Performance Scorecards
- Analytical Capability Development

Case 6. Data-Centre 24/7 Product Screen

References: [1] [4] [30] [5] [2]

Large electricity buyers increasingly care about both when clean energy is produced and how many clean MWh are bought over a year. The seller must separate physical supply, financial settlement, certificate evidence, residual firming, credit, and control rights. The decision is whether a renewable IPP should offer a 24/7-style product, narrow the promise, partner, or walk away.

Decision Question

Should the seller offer hourly matched clean power, a narrower shaped product, a partnered offer, or no-bid?

Situation

A data-centre buyer requests a clean-power product for a flat 40 MW load. The buyer asks for high hourly matching, budget stability, and clear evidence treatment.

The seller has solar assets, access to a battery project, and a trading partner. It does not own enough diverse clean generation to self-supply every hour.

Scenario Data

These are scenario assumptions.

Item	Value
Buyer load	40 MW flat
Annual demand	350.4 GWh/year
Seller-controlled clean supply	260 GWh/year
Annual clean-energy coverage	74%
Hourly matching before extra firming	58%

Item	Value
Target hourly matching for premium product	85%
Residual stress-hour shortfall	22 MWh/h
Stress hours	300 h/year
Firming premium in stress hours	150 EUR/MWh
Buyer premium for stronger hourly matching	4 EUR/MWh

Tasks

1. Calculate the annual buyer demand.
2. Calculate stress-hour firming-at-risk.
3. Compare the buyer premium with the stress exposure.
4. Define the product boundary and evidence standard.
5. Recommend the product boundary.

Worked Solution

Annual demand:

```
40 MW x 8,760 h/year = 350,400 MWh/year
```

Stress-hour firming-at-risk:

```
22 MWh/h x 300 h/year x 150 EUR/MWh
= 990,000 EUR/year
```

Buyer premium:

```
350,400 MWh/year x 4 EUR/MWh
= 1,401,600 EUR/year
```

The premium covers only the illustrated stress-hour procurement cost before collateral, credit, evidence handling, operational costs, margin, and any additional clean supply needed to move from the initial `58%` hourly matching level toward the `85%` threshold. The case still needs careful structuring before rejection or approval.

Product Boundary

The product should be split into claims that can be evidenced and controlled.

Claim Or Service	Required Evidence Or Control	Seller Exposure If Vague
Annual clean-energy coverage	generation volumes, certificates, delivery period, geography, allocation rules	buyer receives a green volume claim but not an hourly product
Hourly matching threshold	hourly load, hourly clean supply, storage dispatch, residual procurement records	seller may be judged against a standard it did not price
Residual firming	cap, index, pass-through, partner supply, or reopener for hard hours	uncapped procurement exposure in scarce hours
Budget stability	settlement reference, volume tolerance, collateral and credit terms	clean-energy product becomes an unpriced financial hedge
Certificate treatment	certificate vintage, cancellation, location, matching period, audit trail	environmental claim can fail after the commercial deal is signed
Buyer flexibility	whether uninterruptible power supply (UPS), cooling, workload shifting, or on-site generation can be used and who controls it	load flexibility is assumed in the model but unavailable in operation

Claim Or Service	Required Evidence Or Control	Seller Exposure If Vague
Reliability boundary	priority between buyer uptime, backup policy, grid services, and seller firming obligations	a flexibility resource is counted in the product when the buyer will not release it

For a data-centre buyer, flexibility resources can be valuable, but reliability constraints come first. Do not count buyer-side flexibility as seller supply unless the buyer grants defined availability, control, telemetry, settlement rights, and an uptime-compatible operating policy.

Numerical Intuition

Annual coverage can look acceptable while hourly matching is weak.

Metric	Result	Why It Matters
Annual clean-energy coverage	74%	Useful annual signal; timing exposure remains
Hourly matching before firming	58%	Shows residual hourly procurement need
Stress-hour firming-at-risk	0.99 million EUR/year	Main seller exposure in hard hours
Buyer premium	1.40 million EUR/year	Commercial headroom before other costs

Decision Memo

Recommendation: offer a partnered hourly-matching product only if the contract defines the evidence standard, residual firming boundary, settlement method, fallback procurement rule, buyer-flexibility rights, and the applicable accounting or certificate framework. Otherwise offer a narrower shaped product and leave the buyer with the residual claim.

The seller should not promise a broad 24/7 clean-energy claim without the applicable accounting and certificate framework in the contract. The commercial product can still be valuable if it says exactly what is being delivered: hourly matched supply to a defined threshold, financial settlement against a defined reference, and residual procurement above a defined boundary.

The recommended offer is therefore a product ladder:

Offer	Seller Commitment	Buyer Benefit	When To Use
As-produced plus certificates	clean energy volume and evidence	lowest complexity	buyer accepts timing exposure
Shaped clean product	defined hourly shape with residual boundary	budget and profile improvement	buyer values shape but not full hourly claim
Partnered hourly-matching product	threshold-based hourly matching with named firming partner and evidence pack	stronger 24/7-style claim	premium covers firming, evidence, credit, and operational cost
No-bid on broad claim	seller rejects uncapped 24/7 language	avoids mispriced obligation	product boundary cannot be made auditable

Assumption Flip

The recommendation changes if stress-hour shortfall rises to 40 MWh/h.

```
40 MWh/h x 300 h/year x 150 EUR/MWh
= 1,800,000 EUR/year
```

That exceeds the buyer premium before other costs. The seller should then require a higher premium, a partner portfolio, a narrower matching threshold, or a no-bid decision.

Failure Mode To Avoid

Offer the product as a premium clean-power contract because annual clean-energy coverage is high, while leaving hourly matching rules, certificate evidence, and residual firming treatment to later documentation.

That decision sells the label before defining the product. Annual matching can hide the hours that drive firming cost, evidence disputes, and residual procurement exposure. A flat load needs a time-indexed product boundary alongside the annual energy total.

Links Back To The Lectures

- Load Is Not Energy
- Market Clock
- Solar Production
- PPA Structure Catalog
- PPA Pricing
- Forecasting
- Portfolio Modeling And Hedging
- Trading Risk, Credit, Liquidity, And Controls
- Future Of PPAs
- Data Centres, 24/7 Procurement, And Flexibility

Case 7. Year-One Energy-Management Operating Model

References: [6] [29] [31] [28] [1] [5]

An independent power producer (IPP) becomes an energy manager when it decides which revenue and risk decisions it must own, which execution activities it can outsource, what raw data it has rights to inspect, and which controls stop a model from becoming an unmanaged production process. This case forces the year-one resource allocation.

Decision Question

Given limited budget, headcount, partner data gaps, and several competing value pools, what should the company build, buy, outsource, audit, or postpone in year one?

Situation

The company has operating solar assets, a growing development pipeline, several structured PPA discussions, and first battery projects entering commercial design. Trading execution and balancing are mostly outsourced. Partner reporting is useful but incomplete. Management wants a credible year-one operating model without approving a large desk or full energy trading and risk management (ETRM) implementation.

The operating model must support four decisions:

- shaped PPA pricing
- BESS sizing and dispatch value
- BRP/BSP and optimizer oversight
- portfolio exposure and risk escalation

Scenario Data

These are scenario assumptions.

Item	Value
Operating renewables	`600 MW`
Development pipeline under commercial review	`1.8 gigawatts (GW)`

Item	Value
First BESS projects	`3 projects`
Annual PPA offers under active discussion	`700 GWh/year`
Outsourced execution volume	`500 GWh/year`
Current internal analytics team	`1 accountable owner + 1 analyst`
Additional hire allowed	`0.5 data engineer` contractor
Year-one budget	`900,000 EUR`
External execution model	BRP/BSP and optimizer partners
Management constraint	no internal trading desk in year one

Partner data-right gaps:

Data Right	Current Status	Operating Consequence
Forecast versions and timestamps	partial	hard to judge what was knowable
Bid and schedule revisions	partial	partner actions cannot be fully reconstructed
Optimizer recommendations	limited	manual override versus model decision is unclear
Meter and settlement files	available with delay	monthly attribution lags
Reserve activation and imbalance detail	inconsistent	revenue stack cannot be reconciled cleanly
Control-right map	contract language exists, operational map missing	dispatch priority conflicts are hard to prove

	Current Status	Operating Consequence
Credit, collateral, and liquidity evidence	fragmented	cannot tell whether feasible actions were blocked by limits or cash timing

Value pools:

Value Pool	Scenario Value At Stake	Evidence Quality
Shaped PPA risk pricing	`560,000 EUR/year`	medium-high
BESS design and revenue stack	`750,000 EUR/year`	medium
Partner leakage reduction	`350,000 EUR/year`	low until data rights improve
Portfolio risk and hedge decisions	`250,000 EUR/year`	medium

Control failures observed:

- commercial offers use inconsistent shape, basis, certificate, and residual-firming assumptions
- BESS cases double-count merchant and PPA use of the same MWh in some sensitivities
- partner reports show results without complete raw files
- management sees portfolio exposure after settlement rather than before decisions
- model versions and assumption owners are unclear

Tasks

1. Allocate the `900,000 EUR` budget across year-one capabilities.
2. Assign headcount ownership and name what remains outsourced.
3. Prioritize a 90-day plan and a 12-month roadmap.
4. State which partner data-right gaps block partner action.
5. Define kill criteria for the roadmap.

Worked Solution

Year-one resource allocation:

Capability	Build, Buy, Outsource, Audit, Or Postpone	Budget	Owner	Reason
Data contracts and reconciliation	build/audit	`130,000 EUR`	accountable owner + data engineer	unlocks all other controls
PPA quote stack and risk map	build	`160,000 EUR`	accountable owner + commercial lead	shaped offers need consistent risk pricing
BESS sizing and dispatch screen	build with optimizer input	`140,000 EUR`	analyst + investment owner	IC decisions are live
Partner data-right remediation	audit/renegotiate	`120,000 EUR`	accountable owner + legal/commercial	raw evidence is missing
Portfolio risk pack	build	`110,000 EUR`	analyst + risk owner	management needs one exposure view
Forecast/vendor evaluation layer	buy inputs, build evaluation	`90,000 EUR`	analyst	value-weighted errors support multiple decisions
Feasible benchmark pilot	build small pilot	`50,000 EUR`	accountable owner	method first, automation later

Capability	Build, Buy, Outsource, Audit, Or Postpone	Budget	Owner	Reason
Contingency and independent review	buy	`100,000 EUR`	steering group	avoids brittle models and weak validation
Total		**`900,000 EUR`**		

Postpone:

- internal trading desk
- full ETRM implementation
- automated benchmark platform across all assets
- proprietary weather model

Outsource:

- BRP/BSP market access
- 24/7 dispatch execution
- base weather/price forecast feeds
- some legal/accounting/certificate advice

Own internally:

- PPA risk allocation and quote logic
- BESS investment assumptions
- data-right requirements
- feasible benchmark methodology
- monthly portfolio risk view
- model register and assumption ownership

Forced Tradeoff

Building a polished partner scorecard immediately would spend money before the evidence layer is ready. Partner data-right gaps make a full scorecard expensive and weak. The stronger year-one allocation funds the data annex and only a narrow benchmark pilot.

```
full_scorecard_without_data_rights = high spend + weak attribution
data_rights_first + small pilot = slower automation + stronger governance
```

The PPA quote stack and BESS screen receive more year-one funding because live commercial and IC decisions can use them with available internal data. Partner commercial action waits for raw evidence.

First 90 Days

Month	Deliverable	Decision It Enables
1	data-rights inventory and gap notice to partners	shows what can be measured, audited, and renegotiated
1	credit, collateral, and liquidity evidence map	shows whether trading constraints are economic, operational, or treasury-driven
1	model register and assumption owner map	prevents uncontrolled spreadsheets
2	shaped-PPA quote stack v1	quote, redesign, no-bid, or pass-through decision
2	BESS duration/revenue-stack screen v1	1h/2h/4h, grid-right, degradation, and control-right decisions
3	first monthly portfolio risk pack	management view of price, shape, basis, imbalance, credit, liquidity, and partner exposure

Month	Deliverable	Decision It Enables
3	partner benchmark methodology note	agrees feasible benchmark rules before automation

The first 90 days should produce usable decision artifacts, not a finished system.

Twelve-Month Roadmap

Quarter	Milestone	Acceptance Test
Q1	common data contract inventory	every critical feed has source, owner, timestamp, unit, granularity, validation rule, and gap owner
Q2	PPA and BESS models used in live decisions	at least one quote or IC recommendation changes because of the model
Q2	risk pack live	management uses it for limits, hedge questions, partner escalation, or deal approval
Q3	forecast evaluation layer	day-ahead, intraday, BESS, and partner forecast errors are value-weighted
Q3	partner data annex amended or escalation logged	missing raw evidence has a commercial owner
Q4	benchmark pilot completed	one asset class or partner has feasible attribution using time-available data

Risk Pack

The recurring risk pack should be short and decision-linked.

Section	Minimum Content	Escalation Trigger
Executive exposure	price, shape, basis, merchant tail, and liquidity summary	exposure outside approved range
PPA pipeline	quote stack, residual firming, certificates, credit, basis	unpriced risk in term sheet
BESS and dispatch	revenue stack, state-of-charge constraints, degradation, reserve/PPA conflicts	double-counted MW/MWh or cycle budget
Partner scorecard	data completeness, forecast timeliness, feasible benchmark gap	raw evidence below requirement
Settlement and reconciliation	meter, imbalance, reserve, invoice exceptions	unreconciled material variance
Credit and collateral	counterparty exposure, collateral calls, guarantees	liquidity stress or credit breach
Model changes	version changes, assumption moves, validation exceptions	unapproved model used in decision
Partner governance	missing files, late explanations, service-level agreement (SLA) breaches, unresolved incident owners	repeated defect without remediation owner

The pack is a control only if management uses it to approve, stop, redesign, hedge, escalate, or request evidence.

Data Rights Before Partner Action

Partner action is blocked where raw evidence is missing.

Action	Evidence Required First
fee reduction or clawback	complete bids, schedules, dispatch recommendations, activations, and settlement data
mandate change	proof that current limits caused avoidable loss or excess risk
optimizer replacement	evidence that feasible recommendations were materially worse than benchmark
BRP/BSP replacement	repeated residual execution gap after data, constraints, forecast, and risk-limit attribution
incentive redesign	clear split between market conditions, constraints, and controllable partner decisions

Until those files exist, the company can demand data remediation and better reporting. It should be cautious about claiming execution leakage.

Kill Criteria

Workstream	Continue If...	Kill Or Redesign If...
PPA quote stack	used in live offers and changes price or structure	commercial team bypasses it or outputs cannot explain residual risk
BESS screen	changes IC decisions or preserves valuable design option	dispatch value cannot be tied to grid rights, degradation, and control rights
Data remediation	partner completeness and timeliness improve against dated milestones	contract rights remain blocked after escalation

Workstream	Continue If...	Kill Or Redesign If...
Risk pack	management takes decisions from it	it becomes a passive report
Forecast layer	value-weighted error improvement exceeds cost	updates arrive too late for action
Benchmark pilot	attribution uses time-available data and settlement reconciliation	it relies on hindsight or missing files

Decision Memo

Recommendation: build a small control-oriented energy-management function in year one while keeping market execution outsourced.

Approve the full `900,000 EUR` allocation shown above. The first year should produce a PPA quote stack, BESS design screen, data-right remediation, monthly portfolio risk pack, forecast evaluation layer, and a narrow feasible-benchmark pilot. Do not approve an internal trading desk or full ETRM implementation until data rights, independent settlement checks, risk limits, and evidence-based partner attribution are working.

Add a monthly energy-risk forum with named authority for limits, credit and collateral exceptions, model approvals, partner escalations, and stop/redesign decisions. A small team can run the operating model only if decision rights are explicit.

The operating model is deliberately constrained:

- one accountable owner coordinates decisions, partner governance, and steering
- one analyst owns models, risk pack production, and value-weighted forecast analysis
- the half-time data engineer builds data contracts, reconciliation checks, and repeatable feeds
- partners continue to execute, but their raw evidence obligations become explicit

The first-year system should be deliberately small but auditable. Every model that affects a quote, dispatch view, partner scorecard, or investment recommendation needs an owner, version, source-data list, change log, and decision record. That minimum control layer should be in place before a larger trading platform or internal desk is considered.

Assumption Flip

The recommendation changes if partners refuse or cannot provide forecast, bid, schedule, dispatch, meter, activation, and settlement files on a reliable timeline.

In that case, shift more budget from model expansion to contract remediation, data interfaces, and replacement-option analysis. Building more models before the company can observe partner decisions would create false control.

Failure Mode To Avoid

Internalize market execution before the company has data rights, independent settlement checks, feasible benchmarks, risk limits, and escalation rules.

That sequence creates operational and control risk. Execution capability only helps when the organization can see the decisions being made, attribute outcomes, enforce limits, and settle disputes with evidence.

Links Back To The Lectures

- Forecasting, Error, And Commercial Value
- Trading Partners And Performance Scorecards
- Analytical Capability Development
- Data And Machine Learning In Energy Management
- Trading Capability Design
- IPP Operating Models For Active Portfolio Management

Glossary

An alphabetical reference for the main terms used in the course.

Actionability factor

Fraction of a model improvement that can become value after information timing, market access, liquidity, control rights, and constraints.

Actionability-adjusted uplift

Raw model or capability uplift after reducing for timing, tradable volume, liquidity, asset constraints, control rights, and organizational adoption.

Activation

Automatic or instructed conversion of reserve capability into physical response.

aFRR

Automatic Frequency Restoration Reserve: reserve automatically activated to restore frequency/control area balance after disturbance.

Aggregator

Entity that combines distributed assets or loads so they can participate commercially or technically as a portfolio.

Allocated capacity

Grid capacity assigned or reserved for a connection or network user under a specific process or agreement.

Analytical engine	
Connected set of models, data flows, controls, and decision rights that turns market and asset information into repeatable commercial decisions.	
Annual matching	
Comparing clean-energy procurement and consumption over a year.	
Apparent power	
Combined active and reactive loading of AC equipment, commonly represented as S where S^2 = P^2 + Q^2.	
As-produced PPA	
PPA where buyer takes generation as produced, inheriting shape/profile exposure.	
Assumption-flip test	
Search for the smallest plausible change in an assumption that would reverse a recommendation.	
Augmentation	
Addition or replacement of battery equipment to restore or maintain usable capacity.	
Back office	
Function responsible for confirmations, settlements, invoicing, contract administration, reporting, and operational control records.	
Backtest	
Historical simulation of model or policy performance using only information available before each decision time.	

Balance obligation

Obligation to plan for balance and financially settle imbalances under market rules.

Balancing capacity

MW held available for balancing service under TSO product rules, whether or not activation occurs.

Balancing energy

Energy delivered or absorbed when balancing bids or reserves are activated.

Balancing market

Real-time or near-real-time mechanism used by the TSO to maintain system balance.

Bankable revenue

Revenue treated as dependable enough for financing, depending on contract, counterparty, tenor, and downside protection.

Baseload PPA

PPA with a flat delivery profile over a defined period.

Basis risk

Mismatch between the exposure being hedged and the instrument/reference used to hedge it.

Battery duration

Battery energy capacity divided by power rating, usually expressed in hours.

Behavioral market power

Actual conduct that uses market-power potential through bidding, withholding, scheduling, or dispatch choices.

BESS

Battery Energy Storage System.

BRP

Balance Responsible Party: entity financially responsible for imbalances at a delivery point or portfolio.

BSP

Balance Service Provider: entity providing balancing services to the TSO.

C-rate

Charge or discharge rate relative to battery energy capacity; 1C roughly means full usable charge or discharge in one hour.

Calculation period

Time period over which a reference price or settlement amount is calculated.

Calendar aging

Battery degradation associated with time, temperature, and state-of-charge conditions, even without heavy cycling.

Cannibalization

Value erosion when similar generators produce at the same time and depress prices during their own production hours.

Cap

Contractual maximum price or value above which upside is limited or shared, depending on structure.

Capability priority score

Heuristic ranking of a capability by value at stake, controllability, observability, urgency, cost, and risk.

Capability-contract matrix

Build/buy/outsource table that separates internal judgment, tools, outsourced services, and non-negotiable control/data rights.

Capacity credit

Dependable capacity contribution of a resource divided by its nameplate capacity.

Capacity factor

Actual energy produced over a period divided by energy that would be produced at full nameplate output over that period.

Capacity remuneration mechanism

Market or regulatory mechanism that pays for capacity, availability, or reliability contribution in addition to energy.

Capacity warranty

Warranty or performance guarantee specifying retained usable battery capacity over time, usually conditional on operating conditions and measurement method.

Capture price

Generation-weighted average market price received by an asset or technology.

Capture rate	
Capture price divided by average spot price for the same market/period.	

Capture rate

Capture price divided by average spot price for the same market/period.

Cash available for debt service

Cash flow available to pay scheduled debt service after operating revenue, operating costs, and allowed deductions/reserves according to the model or financing terms.

Cash Flow at Risk

Liquidity metric focused on potential cash demands versus available sources over a horizon.

Certificate contract price

Price or embedded value assigned to certificates such as GOs or EECS certificates in a PPA.

CfD

Contract for Difference: financial contract settling the difference between a strike/reference price and a market/reference price.

Closed-loop latency ratio

Feedback latency divided by decision cadence; high values mean the organization learns slower than it acts.

Collateral

Value posted to reduce credit exposure between counterparties or clearing participants.

Congestion rent

Value associated with a price difference across a constrained transmission interface multiplied by feasible flow, in simplified market-design examples.

Connection agreement

Agreement governing how an asset connects to the grid, including technical, capacity, cost, and operational conditions.

Connection charge

Charge, often one-off, covering part or all of connecting a new user or upgrading a connection.

Constraint

Physical, market, contractual, or operational limit a model or decision must obey.

Construction contribution

One-off payment or cost contribution for grid works required to connect or serve a network user.

Contract quantity

Electricity quantity specified by a PPA or related contract, such as all metered output, part of metered output, or a fixed delivery schedule.

Control right

Authority to decide, instruct, or veto scheduling, bidding, dispatch, curtailment, hedging, or reserve actions.

Cost of capital

Required return for the capital supplied to a project, including debt and equity costs.

Counterparty credit risk

Risk that the counterparty to a contract does not pay, deliver, or otherwise perform as required.

Credit support

Collateral, guarantee, parent support, letter of credit, or similar instrument used to reduce counterparty credit risk.

Cross-border PPA

PPA where buyer and generation asset are in different national markets or bidding zones, often creating basis and transmission-right questions.

Curtailment

Reduction of output below available generation, for economic, grid, or system-flexibility reasons.

Customer value

Buyer's benefit from a product, such as hedge value, budget certainty, decarbonization claim, hourly alignment, or procurement simplicity; distinct from seller cost.

Customer willingness-to-pay

Maximum price or premium a buyer may rationally accept based on hedge value, budget certainty, decarbonization value, evidence needs, and procurement alternatives.

Cycle aging

Battery degradation associated with charge-discharge use, including throughput, depth of discharge, rate, and operating pattern.

Daily risk pack

Recurring report connecting positions, MtM, market risk, credit exposure, liquidity/collateral, basis/shape, partner execution, limits, data quality, and actions.

Data contract

Definition of a data feed's source, owner, timestamp, unit, granularity, validation rule, and decision use.

Data right

Right to receive data needed to reproduce, audit, and govern decisions.

Day-ahead market

Auction before delivery day where hourly or sub-hourly electricity volumes are cleared for the next day.

Debt

Money lent to a project or company, usually paid before equity and therefore usually cheaper but less tolerant of downside cash-flow weakness.

Debt sculpting

Shaping debt service over time to fit the cash-flow profile a lender is willing to count.

Debt sizing

Process of estimating how much debt a project can support from lender-counted cash flow, coverage requirements, tenor, interest rate, reserves, and amortization structure.

Decision latency

Time between the arrival of useful information and the latest moment a feasible action can still be taken.

Decision memo

Short recommendation document stating the decision, core evidence, retained risk, binding assumptions, owner, next measurement, and assumption-flip test.

Decision owner

Person or function accountable for accepting, rejecting, or overriding a model recommendation or operational action.

Decision variable

Quantity the optimization model is allowed to choose.

Degradation

Loss of battery capacity or performance over time due to calendar aging and cycling.

Degradation-adjusted spread

Discharge price minus charging cost adjusted for efficiency, degradation, fees, tariffs, and opportunity cost.

Delivery schedule quantity

Fixed quantity or quantities to be delivered or settled during defined periods under a contract schedule.

Depth of discharge

Fraction of stored energy practically used in a discharge.

Dispatch

Operational choice of when/how much an asset charges, discharges, generates, or bids.

Dispatchable

Controllable by an operator within technical and fuel/resource constraints.

Distribution

Lower-voltage network delivery between the transmission system and end users / distributed resources.

Diversification

Risk reduction from imperfectly correlated exposures rather than asset count alone.

DSCR

Debt service coverage ratio: cash available for debt service divided by required debt service over a period.

DSO

Distribution System Operator: entity responsible for operating a distribution network.

Duration hedge

Extra battery energy capacity bought partly to handle uncertainty in spreads, delivery obligations, reserve calls, curtailment windows, or grid constraints.

Economic withholding

Offering capacity at a price high enough that it is unlikely to clear, even though it is technically available.

Energy

Quantity accumulated or consumed over time, usually Wh, kWh, MWh, TWh.

Energy management

Capability connecting forecasts, market positions, dispatch, contract obligations, imbalance exposure, partner execution, and portfolio risk.

Energy-limited

Able to provide power only until a stored-energy, fuel, water, or demand-flexibility constraint binds.

Energy-management operating system

Set of recurring decisions, data flows, models, limits, partner controls, feedback loops, and review rhythms that turns assets and contracts into governed commercial action.

Energy-only market

Market design where investment recovery relies mainly on energy and ancillary-service revenues, including scarcity prices.

Equity

Ownership capital that receives residual cash flows after operating costs, debt service, reserves, and other senior claims.

Equivalent full cycle

One full charge-discharge equivalent measured by throughput relative to usable energy capacity; partial cycles can add up to equivalent full cycles depending on the convention used.

ETRM

Energy trading and risk management system used to record, value, monitor, and control trades, positions, limits, settlement, and reporting.

Expected shortfall

Average loss conditional on losses exceeding a selected tail threshold such as VaR.

Expected unserved energy

Expected MWh of demand not served because supply is insufficient.

Exposure cube

Portfolio map that classifies risk by geography, technology, contract, hedge, flexibility, counterparty, liquidity, and data/control rights.

Fault level

Measure related to system strength and available fault current at a network point.

FCR

Frequency Containment Reserve: reserve intended to contain frequency deviations quickly after an imbalance.

Feasible counterfactual

Simulated action path that could have been taken with contemporaneous information, constraints, market access, risk limits, and control rights.

Feedback latency

Time between a decision and the reliable measurement or settlement data needed to evaluate it.

Final position

Market position carried into delivery after accepted trades, nominations, or adjustments.

Financeability lens

View of project cash flows focused on resilient cash for debt and covenant/downside requirements alongside equity upside.

Firming cost

Cost of transforming variable renewable output into a firmer or more useful delivery shape through storage, portfolio netting, market purchases, hedging, or flexibility.

Firming-at-risk

Stress measure for the cost of residual firming exposure: residual MWh in hard hours multiplied by the relevant firming premium or stress price.

Flexibility potential

Technical or practical ability of a load, generator, storage asset, or portfolio to change injection/withdrawal in response to grid or market needs.

Flexible connection agreement

Connection or network-use arrangement that allows limited, conditional, or interruptible injection or withdrawal, often to accelerate connection or reduce reinforcement needs.

Floor

Contractual minimum price or value below which settlement is protected, depending on structure.

Forecast-based dispatch

Dispatch optimized using information available at decision time.

FPA	Flexibility Purchase Agreement or flexibility-style arrangement; exact meaning must be defined by source/context before use.
Front office	Commercial function that originates, structures, prices, trades, dispatches, or manages market-facing portfolio decisions.
Generation adequacy	Generation-resource component of system adequacy.
Governance calendar	Recurring cadence for position review, partner performance, PPA pricing, portfolio exposure, model assumptions, capability strategy, and investment gates.
Granular certificate	Certificate or attribute-tracking concept that adds temporal and sometimes locational granularity to energy attribute tracking.
Grid tariff	Charge paid for use of the grid, potentially including fixed, capacity, energy, loss, and reactive-power components.
Grid-following inverter	Inverter that synchronizes to an existing grid voltage/frequency reference.
Grid-forming inverter	

Inverter controlled to establish or support voltage/frequency reference behavior more actively.

Hedge-quality checklist

Test of whether a hedge matches the underlying exposure by reference price, volume, shape, tenor, basis, liquidity, collateral, and governance.

Hourly matching

Comparing clean supply and consumption within each hour.

Imbalance

Difference between scheduled/nominated and actual generation or consumption.

Incremental duration

Additional MWh added to a battery design while keeping MW approximately fixed.

Incremental duration value

Additional value from adding MWh to a battery after additional revenue, capex, degradation, tariffs, and opportunity cost.

Independent price validation

Process of checking marks, curves, or valuations independently from the team that originated or owns the position.

Inertia

Resistance of rotating mass to sudden frequency change.

Insurance operating covenant

Condition in an insurance policy or endorsement that limits how an asset may be operated while keeping coverage, pricing, or claim eligibility intact. For BESS this may include cycle, throughput, duty-cycle, safety, monitoring, or approved-use restrictions.

Intraday market

Market after day-ahead clearing where participants adjust positions closer to delivery.

Investment-case waterfall

Path from gross physical value to equity cash flow that shows capture, curtailment, contract, grid, BESS, opex, debt, and residual effects layer by layer.

IPP

Independent Power Producer: company that owns or operates generation assets and sells electricity or related products without being the incumbent utility.

IRR

Internal rate of return: discount rate that makes project NPV equal zero.

Limited grid access

Connection or access arrangement where injection or withdrawal is constrained under specified conditions.

Liquidity at Risk

Risk measure for liquidity pressure under market, credit, collateral, or trading-liquidity stress.

Liquidity risk

Risk that a firm cannot trade, exit, or fund positions when needed without unacceptable cost or disruption.

Load	
Electrical demand at a point, portfolio, zone, or system over time.	
Load shape	
Pattern of demand across time.	
Long-duration energy storage	
Storage intended to address longer scarcity periods than short intra-day shifting.	
Loss charge	
Charge or adjustment intended to recover electrical losses associated with network use.	
Loss of load expectation	
Expected duration of insufficient supply over a planning horizon.	
Loss of load probability	
Probability that available resources cannot meet load in a specified period.	
Loss tree	
Engineering model that tracks energy losses through a renewable, storage, or hybrid system so dispatch and valuation use deliverable energy rather than ideal nameplate energy.	
Margin call	
Demand to post additional collateral after exposure or collateral requirements change.	

Marginal duration hurdle	Annualized cost and opportunity-cost threshold that the next MWh of storage energy capacity must clear before extra duration is worth buying or building.
Marginal risk contribution	Incremental stress, variance, liquidity, or concentration added by the next asset, hedge, contract, or partner dependency.
Market access	Ability to participate in a market through required contracts, systems, qualifications, collateral, and operations.
Market monitoring	Monitoring of bids, outages, constraints, behavior, and outcomes to distinguish normal scarcity, opportunity cost, and problematic conduct.
Market power	Ability to profitably influence market price away from the competitive level for a meaningful period.
Market risk	Exposure to losses from movements in prices, spreads, volatility, basis, or correlations.
Market time unit	Time granularity of traded or settled energy, such as hourly or 15-minute periods depending on market and date.
Merchant tail	

Post-contracted or partially uncontracted period where project revenue depends more directly on market prices and exposed risks.

Merchant upside

Uncontracted or weakly contracted expected value from market optimization.

mFRR

Manual Frequency Restoration Reserve: reserve manually or centrally activated to restore balance and manage system needs.

Middle office

Independent or semi-independent function that measures risk, validates prices, monitors limits, benchmarks performance, and challenges front-office decisions.

Missing money

Concern that market revenues may be insufficient to finance needed capacity because scarcity or investment signals are muted, risky, capped, or incomplete.

Mixed-integer linear programming

Optimization method where some variables are continuous and others are integer or binary, useful when dispatch choices include on/off, mutually exclusive, or commitment-style decisions.

Model inventory

Register of models, owners, purposes, inputs, outputs, limits, validation status, and retirement criteria.

Model register

Inventory of models, owners, inputs, outputs, validation status, and decisions supported.

Model value at risk

Scenario estimate of value exposed to a wrong model-driven action across affected MW, duration, adverse price error, and bad-event count.

Model-risk charge

Scenario or governance allowance for losses, overrides, monitoring, and control burden introduced by relying on a model.

Multi-buyer PPA

PPA where several buyers aggregate demand to contract with one generator or supplier structure.

Nameplate capacity

Rated maximum power under defined conditions.

NEMO

Nominated Electricity Market Operator: exchange/operator designated for day-ahead and intraday market coupling functions.

Netting

Partial cancellation of positive and negative deviations inside compatible commercial or settlement boundaries.

NPV

Net present value: discounted value of future cash flows minus investment cost.

Numerical weather prediction

Physics-based weather model used as an input to power, load, price, and system forecasts.

Objective function

Mathematical expression a model seeks to maximize or minimize.

Offtaker

Buyer of electricity or environmental attributes under a commercial contract such as a PPA.

Opportunity cost

Value of the best feasible alternative action forgone when an asset is committed to a different action.

Paid-or-required capability gap

Difference between system-service capability that is required or valuable at a site and capability that is already qualified and available.

Partner benchmark

Feasible counterfactual used to compare actual BRP/BSP/optimizer performance with actions available under real constraints and rights.

Pay-as-produced PPA

PPA where the buyer purchases electricity as generated, so the buyer generally takes more generation-profile exposure.

Peak load

Maximum load over a defined period and geography.

Perfect foresight dispatch	Dispatch optimized with actual future information; useful as an upper bound for operational benchmarks built from tradable information available at decision time.
Physical withholding	Making capacity unavailable even though it could physically produce.
Pivotal supplier	Supplier whose capacity is needed to meet demand under the relevant constraints.
Portfolio exposure	Sensitivity of portfolio value or cash flow to a market, physical, contractual, credit, or operational driver.
Portfolio PPA	PPA backed by a portfolio of assets rather than a single identified plant, potentially spreading resource and operational risks.
Potential future exposure	Forward-looking estimate of how large counterparty exposure may become under future market movements.
Power	Rate of energy flow, usually measured in W, kW, MW, or GW.
PPA	Power Purchase Agreement: contract for purchase/sale of electricity, often long-term.

Private-wire PPA	
PPA where generation and consumption are connected through a direct/private wire rather than only through ordinary grid supply arrangements.	
Product-readiness score	
Heuristic assessment of whether a firm has the assets, portfolio, data, control rights, risk limits, and evidence needed to offer a more advanced PPA product.	
Project policy constraint	
Project-specific insurance, lender, warranty, or operating-policy limit on cycling, throughput, duty cycle, approved revenue modes, data retention, or claim eligibility.	
Qualification chain	
Sequence that turns technical capability into commercial value: specify, test, qualify, control, measure, settle, and enforce.	
Quote stack	
Pricing structure that decomposes an offered PPA price into raw energy, shape/firming, imbalance, grid/tariff, certificates, credit, tenor, optionality, and risk-margin components.	
Reactive power	
Component of AC power associated with voltage support and electric/magnetic fields, commonly measured in var/MVAr.	
Reference price	
Market price, index, or formula used to settle a financial PPA, CfD, floor/cap, or indexed price component.	

Regional upscaling	Converting asset-level forecasts or samples into a portfolio, bidding-zone, or control-area forecast.
Reserve capacity	MW held available for possible TSO activation under product rules.
Reserve headroom	MW or MWh capability held back so a battery can deliver a reserve obligation if called.
Residual demand	Demand left for a supplier after subtracting other available supply and feasible imports.
Residual load	Load remaining after subtracting variable renewable generation or another specified supply category.
Residual obligation	Hourly MWh, EUR, evidence, or operational exposure left after contracted clean supply, storage, hedges, certificates, and control actions have been applied.
Residual Supply Index	Market-power screen comparing supply available without a supplier to demand; values below one indicate pivotality in the simple form.
Retained-risk table	Decision-memo table naming risks that remain after the recommendation, their owners, warning signals, and mitigations.

Revenue quality ladder	Course tool ranking modeled value by how confidently it can support fixed obligations versus equity upside or unproven optionality.
Revenue stack	Set of revenue sources a flexible asset may pursue, subject to mutual exclusivity, opportunity cost, and operational constraints.
Risk limit	Quantitative or qualitative boundary on positions, losses, exposures, stress losses, collateral needs, or permitted strategies.
Round-trip efficiency	Energy discharged divided by energy charged over a cycle.
Scarcity duration	Time length of the shortage or obligation a resource must cover.
Scarcity pricing	High pricing that reflects real scarcity of energy, capacity, flexibility, or network capability.
Scarcity-duration distribution	Distribution showing how often scarcity or obligation events of different lengths occur, for example 1-hour, 4-hour, overnight, multi-day, or seasonal events.
Security of supply	Broad ability of the power system to serve demand reliably across operating and planning horizons.

Shadow price

Marginal value of relaxing a binding constraint by one unit in an optimization problem.

Shadow trading

Counterfactual simulation of feasible trading/dispatch decisions used to benchmark actual execution or partner performance.

Shape cost

Cost of converting asset output into a promised contract profile, including market purchases, storage losses, degradation, hedging, balancing, grid/tariff costs, and risk margin.

Shape risk

Risk that production or consumption timing differs from the contracted or hedged delivery shape.

Short-circuit ratio

Screening concept comparing grid strength to connected converter capacity; detailed definition depends on method.

State of charge

Battery energy currently stored, usually expressed as MWh or percent of usable capacity.

Stress test

Scenario analysis of severe but plausible or intentionally challenging market, credit, liquidity, or operational conditions.

Structural market power

Market-power potential arising from market structure, location, ownership, concentration, constraints, or pivotality.

Subscribed capacity

Capacity level contracted for grid use or tariff purposes.

System adequacy

Longer-term ability of the system to have enough resources and network capability to meet demand under credible conditions.

System margin

Difference between available dependable resources and demand or required reserve/adequacy target, depending on definition.

System security

Short-term ability of the power system to remain balanced and withstand disturbances.

System-service option

Capability that may create value through connection approval, compliance, avoided curtailment, paid services, or future grid requirements, but is not yet base-case revenue.

Terminal value

Value assigned to the final state of a storage model, often to avoid artificial emptying at the end of the optimization horizon.

Termination amount

Amount due on early contract termination, often linked to mark-to-market or alternative valuation methods.

Throughput budget

Allowed or planned cumulative charged/discharged energy over a period or asset life.

Transmission

High-voltage bulk power transport over the main grid.

TSO

Transmission System Operator: entity responsible for secure operation of the transmission system and system balance within its mandate.

Use-of-network charge

Recurring charge for grid use, potentially covering infrastructure, losses, system services, metering, or reactive charges depending on tariff design.

Value at Risk

Loss threshold for a specified probability and horizon; it does not describe the severity of losses beyond the threshold.

Value of lost load

Estimate of consumer willingness to pay to avoid involuntary interruption of electricity supply.

Value pool

Source of possible economic improvement, such as better pricing, reduced imbalance, improved dispatch, or reduced leakage.

Value-duration curve

Curve or table showing the marginal value of adding each extra hour of usable storage duration after efficiency, degradation, grid, tariff, contract, and opportunity-cost constraints.

Value-weighted error

Forecast or model error weighted by the economic consequence of being wrong in that interval or scenario.

Virtual battery

Controllable demand or process flexibility that can absorb, defer, or release electricity demand over time in a way that mimics some storage services without storing electricity electrochemically.

Voltage

Electrical potential difference; grid voltage must remain within operational limits.

WACC

Weighted average cost of capital: combined cost of debt and equity after weighting by capital structure and tax treatment where applicable.

Warranty envelope

Operating conditions under which a battery's warranty, performance guarantee, availability commitment, or remedy structure remains valid.

Weather-dependent

Output materially depends on weather conditions outside operator control.

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